FET small-signal I

Answers and worked solutions · Day 18 · Tue 10 Nov · pack E11

5. Practice

From your course files

WS8 Q7 (course question).

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(12\text{ mA})}{4\text{ V}} = \mathbf{6}\text{ mS}$$

WS8 Q8 (course question).

At $V_{GS} = 0$, Shockley gives $I_D = I_{DSS}$. Rearrange $g_{m0} = 2I_{DSS}/|V_P|$:

$$I_{DSS} = \frac{g_{m0}|V_P|}{2} = \frac{(5\text{ mS})(4\text{ V})}{2} = \mathbf{10}\text{ mA}$$

WS8 Q9 (course question).

The device alone (no R_D):

$$Z_o(\text{FET}) = r_d = \frac{1}{g_{os}} = \frac{1}{25\ \mu\text{S}} = \mathbf{40}\text{ k}\mathbf{\Omega}$$

$$A_v(\text{FET}) = -g_m r_d = -(4.5\text{ mS})(40\text{ k}\Omega) = \mathbf{-180}$$

This is the largest gain the device could ever give (the limit $R_D \to \infty$). Any real R_D gives less.

WS8 Q10 (course question).

READ THE FIGURE. The curve meets the I_D axis at 10 mA, so $I_{DSS} = 10$ mA. It meets the V_{GS} axis at −5 V, so $V_P = -5$ V.

(a) $g_{m0} = \dfrac{2(10\text{ mA})}{5\text{ V}} = \mathbf{4}\text{ mS}$.

(b) Graphically. Draw the tangent at $V_{GS} = -0.5$ V. A clean way to do it from the plotted dots is to take the chord from $V_{GS} = -1$ V ($I_D \approx 6.4$ mA, the dot just below 7) to $V_{GS} = 0$ ($I_D = 10$ mA). That chord is symmetric about −0.5 V:

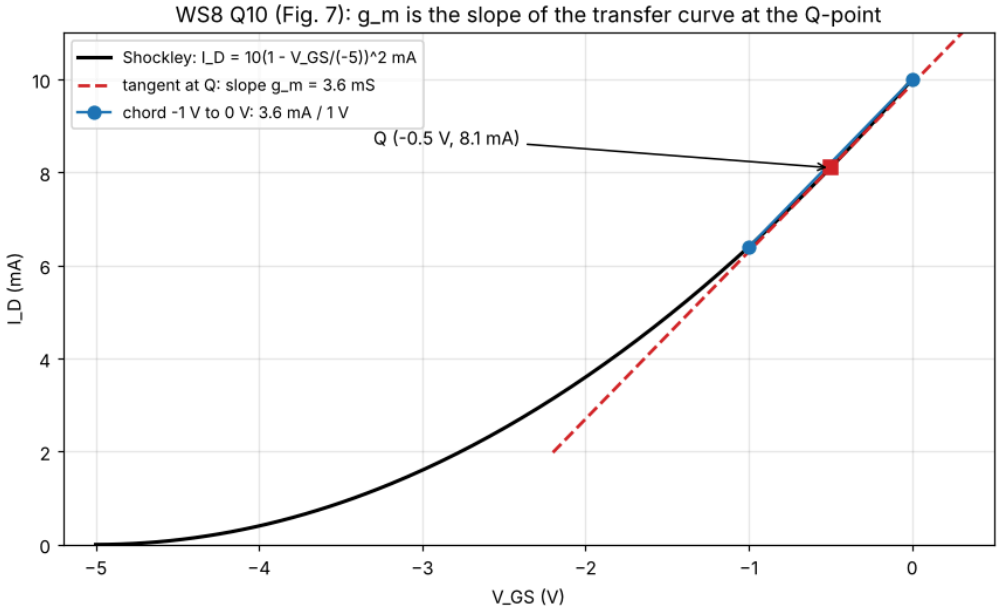
$$g_m \approx \frac{\Delta I_D}{\Delta V_{GS}} = \frac{10 - 6.4\ \text{mA}}{1\ \text{V}} \approx \mathbf{3.6}\ \text{mS}$$

(A hand-drawn tangent will give roughly 3.4–3.8 mS.)

(c) Calculated.

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right) = 4\ \text{mS}\left(1 - \frac{-0.5}{-5}\right) = 4(0.9) = \mathbf{3.6}\ \text{mS}$$

Comparison: the two agree. For a parabola, a chord symmetric about a point has exactly the tangent slope there. A drawn tangent may differ by about ±5 %.



WS8 Q10: tangent and symmetric chord at the Q-point

WS8 Q11 (course question).

Circuit (Fig. 8): $V_{DD} = +18$ V, $R_D = 1.8$ kΩ, $R_G = 1$ MΩ returned to a 1.5 V battery with its − terminal at the gate side. The source is grounded.

DC: $I_G = 0$, so $V_{GS_Q} = -1.5$ V.

$$I_{D_Q} = 10\left(1 - \frac{-1.5}{-6}\right)^2 = 5.625\ \text{mA}, \qquad V_{DS} = 18 - (5.625)(1.8) = 7.88\ \text{V}\ (\text{saturation, since } > 4.5\ \text{V})$$

g_m: $g_{m0} = 2(10)/6 = 3.33$ mS, so $g_m = 3.33(1 - 0.25) = 2.5$ mS.

ac:

- $Z_i = R_G = \mathbf{1}\ \mathbf{M\Omega}$
- $Z_o = R_D \parallel r_d = 1.8 \parallel 40 = \mathbf{1.72}\ \mathbf{k\Omega}$

- $A_v = -g_m(R_D \parallel r_d) = -(2.5\ \text{mS})(1.722\ \text{k}\Omega) = \mathbf{-4.31}$ (−4.5 ignoring r_d)

WS8 Q12 (course question).

Circuit (Fig. 9): $V_{DD} = 12$ V, $R_D = 3.3$ kΩ, $R_G = 10$ MΩ, $R_S = 1.1$ kΩ bypassed by $C_S = 20$ µF.

You do not need a DC analysis here, because $g_m = g_{fs}$ is given directly. R_S is bypassed, so the source is at ac ground.

- $r_d = 1/50\ \mu\text{S} = 20$ kΩ. Note that r_d is only 6× R_D, so you must keep it.
- $Z_i = R_G = \mathbf{10\ \text{M}\Omega}$
- $Z_o = R_D \parallel r_d = 3.3 \parallel 20 = \mathbf{2.83\ \text{k}\Omega}$

(Not asked, but for completeness: $A_v = -(3\ \text{mS})(2.83\ \text{k}\Omega) = -8.50$.)

WS8 Q13 (course question).

Circuit (Fig. 10): $V_{DD} = 20$ V, $R_1 = 82$ MΩ, $R_2 = 11$ MΩ, $R_D = 2$ kΩ, $R_S = 610\ \Omega$ bypassed by C_S. $I_{DSS} = 12$ mA, $V_P = -3$ V, $r_d = 50$ kΩ.

DC:

$$V_G = 20 \cdot \frac{11}{82 + 11} = 2.366\ \text{V}, \qquad V_{GS} = 2.366 - 610\, I_D$$

Substitute into $I_D = 12\ \text{mA}(1 + V_{GS}/3)^2$ and solve. The valid root is $V_{GS_Q} = -0.974$ V, $I_{D_Q} = 5.47$ mA. (A graphical solution gives about −0.95 V and 5.5 mA.)

Check: $V_S = 3.34$ V, $V_D = 20 - 5.47(2) = 9.05$ V, $V_{DS} = 5.71$ V $> V_{GS} - V_P = 2.03$ V, so the device is in saturation.

g_m: $g_{m0} = 2(12)/3 = 8$ mS, so $g_m = 8\left(1 - \frac{-0.974}{-3}\right) = 5.40$ mS.

ac (R_S bypassed):

- $Z_i = R_1 \parallel R_2 = 82 \parallel 11 = \mathbf{9.70\ \text{M}\Omega}$
- $Z_o = R_D \parallel r_d = 2 \parallel 50 = \mathbf{1.92\ \text{k}\Omega}$
- $A_v = -(5.403\ \text{mS})(1.923\ \text{k}\Omega) = -10.39$
- $V_o = A_v V_i = -10.39 \times 20\ \text{mV} = \mathbf{-208}\ \text{mV}$ (about 0.21 V peak, inverted)

(Ignoring r_d gives −216 mV.)

WS8 Q14 (course question).

Circuit (Fig. 11): $V_{DD} = 16$ V, $R_D = 1.1$ kΩ, $R_G = 10$ MΩ from gate to ground, and the source grounded. n-channel D-MOSFET with $I_{DSS} = 8$ mA, $V_P = -3$ V.

DC: the gate is at 0 V (no gate current through R_G) and the source is at 0 V, so $V_{GS_Q} = 0$ V and $I_{D_Q} = I_{DSS} = 8$ mA. $V_{DS} = 16 - 8(1.1) = 7.2$ V > 3 V, so the device is in saturation.

g_m: $V_{GS_Q} = 0$, so $g_m = g_{m0} = 2(8)/3 = 5.33$ mS.

ac: $r_d = 1/20\ \mu\text{S} = 50$ kΩ, $R_D \parallel r_d = 1.076$ kΩ.

$$A_v = -(5.333\ \text{mS})(1.076\ \text{k}\Omega) = -5.74, \qquad V_o = -5.74 \times 2\ \text{mV} = \mathbf{-11.5}\ \text{mV}$$

(−11.7 mV ignoring r_d.) Also, $Z_i = 10$ MΩ and $Z_o = 1.08$ kΩ.

WS8 Q15 (course question).

Circuit (Fig. 12): $V_{DD} = 30$ V, $R_1 = 40$ MΩ, $R_2 = 10$ MΩ, $R_D = 3.3$ kΩ, $R_S = 1.2$ kΩ bypassed. n-channel E-MOSFET with $V_{GS(Th)} = 3$ V, $k = 0.4 \times 10^{-3}$ A/V².

DC:

$$V_G = 30 \cdot \frac{10}{50} = 6\ \text{V}, \qquad V_{GS} = 6 - 1200\, I_D, \qquad I_D = 0.4 \times 10^{-3}(V_{GS} - 3)^2$$

Substitute the second equation into the first: $0.48(V_{GS} - 3)^2 + V_{GS} - 6 = 0$. The roots are $V_{GS} = 4.667$ V and −0.75 V. Reject −0.75 V, because it is below $V_{GS(Th)}$ and the device would be off.

So $V_{GS_Q} = 4.67$ V and $I_{D_Q} = 0.4(1.667)^2 = 1.11$ mA. $V_{DS} = 30 - 1.111(4.5) = 25.0$ V $> V_{GS} - V_{GS(Th)} = 1.67$ V, so the device is in saturation.

g_m **(E-MOSFET formula):** $g_m = 2k(V_{GS_Q} - V_{GS(Th)}) = 2(0.4 \times 10^{-3})(1.667) = 1.33$ mS.

ac:

- $Z_i = 40 \parallel 10 = 8$ MΩ
- $Z_o = 3.3 \parallel 40 = 3.05$ kΩ
- $A_v = -(1.333\ \text{mS})(3.048\ \text{k}\Omega) = -4.06$
- $V_o = -4.06 \times 0.8\ \text{mV} = \mathbf{-3.25}\ \text{mV}$

New questions

N1 [3 marks] (easy).

a. $g_{m0} = 2(8)/4 = \mathbf{4}$ mS.

b. $g_m = 4\left(1 - \frac{-1.5}{-4}\right) = 4(0.625) = \mathbf{2.5}$ mS.

c. $g_m = g_{m0}\sqrt{I_D/I_{DSS}} = 4\sqrt{2/8} = \mathbf{2}$ mS. Check: $I_D = 2$ mA corresponds to $V_{GS} = -4(1 - 0.5) = -2$ V, and $4(1 - 0.5) = 2$ mS.

N2 [4 marks] (design).

DC: $I_{D_Q} = 12(1 - 2/6)^2 = 5.33$ mA. $g_{m0} = 2(12)/6 = 4$ mS, so $g_m = 4(1 - 1/3) = 2.67$ mS.

a. $R_D \parallel r_d = 6/2.667\ \text{mS} = 2.25$ kΩ. Then $R_D = \dfrac{1}{1/2.25\text{k} - 1/60\text{k}} = \mathbf{2.34\ k\Omega}$. The nearest standard value is 2.4 kΩ, which gives $A_v = -6.15$. (2.2 kΩ would give −5.66.)

b. $V_{DS} = 24 - (5.33\ \text{mA})(2.34\ \text{k}\Omega) = \mathbf{11.5}$ V (11.2 V with 2.4 kΩ). This is greater than $V_{GS} - V_P = 4$ V, so the device is in saturation.

N3 [8 marks] (multi-part, chained).

a. $V_{GS} = -680\, I_D$ with $I_D = 6(1 + V_{GS}/3)^2$ mA. The valid root is $V_{GS_Q} = \mathbf{-1.304}$ V, $I_{D_Q} = \mathbf{1.92}$ mA. Then $V_{DS} = 18 - 1.918(3.3 + 0.68) = \mathbf{10.37}$ V, which is above $V_{GS} - V_P = 1.7$ V, so the device is in saturation.

b. $g_{m0} = 2(6)/3 = 4$ mS, so $g_m = 4\left(1 - \frac{-1.304}{-3}\right) = \mathbf{2.26}$ mS. $r_d = 1/20\ \mu\text{S} = \mathbf{50\ k\Omega}$.

c. $Z_i = \mathbf{1\ M\Omega}$, $Z_o = 3.3 \parallel 50 = \mathbf{3.10\ k\Omega}$, $A_v = -(2.261)(3.096) = \mathbf{-7.00}$.

d. $A_v = -\dfrac{(2.261)(3.3)}{1 + (2.261)(0.68)} = \mathbf{-2.94}$ (−2.85 with r_d included). The gain falls by a factor of 2.4.

e. $A_v = -g_m(r_d \parallel R_D \parallel R_L) = -2.261\,(50 \parallel 3.3 \parallel 4.7\ \text{k}) = -4.22$. Then

$$A_{vs} = \frac{1\ \text{M}}{1\ \text{M} + 600}(-4.22) = \mathbf{-4.22}$$

The FET's huge Z_i makes R_{sig} negligible, while the load costs about 40 % of the gain.

N4 [5 marks].

a. $V_G = 20(20/120) = 3.33$ V and $V_{GS} = 3.33 - 330 I_D$. With Shockley, the valid root is $V_{GS_Q} = \mathbf{+0.293}$ V, $I_{D_Q} = \mathbf{9.21}$ mA (more than I_{DSS}, as expected for positive

V_{GS}). $V_{DS} = 20 - 9.214(1.33) = \mathbf{7.75}$ V, which is above $V_{GS} - V_P = 4.29$ V, so the device is in saturation.

b. $g_{m0} = 2(8)/4 = 4$ mS, so $g_m = 4\left(1 - \frac{0.293}{-4}\right) = \mathbf{4.29}$ mS. That is 7 % above g_{m0}, because $V_{GS_Q} > 0$.

c. $Z_i = 100 \parallel 20 = \mathbf{16.7}$ MΩ, and $A_v = -g_m R_D = \mathbf{-4.29}$.

(Design note: with $R_D = 1.5$ kΩ the drain would drop to $V_{DS} = 3.1$ V, below 4.29 V. The device would then be ohmic and the small-signal model would not apply. Always do the check.)

N5 [6 marks] (hard).

a. $V_G = 24(6/24) = 6$ V, $V_{GS} = 6 - 1000 I_D$, $I_D = 0.5\,\text{mA}(V_{GS} - 2)^2$. Substituting gives $0.5(V_{GS} - 2)^2 + V_{GS} - 6 = 0$, so $V_{GS} = 4$ V or −2 V. Reject −2 V. So $V_{GS_Q} = \mathbf{4}$ V, $I_{D_Q} = \mathbf{2}$ mA, and $V_{DS} = 24 - 2(5.7) = \mathbf{12.6}$ V (above 2 V, so saturation).

b. $g_m = 2k(V_{GS_Q} - V_{GS(Th)}) = 2(0.5)(2) = \mathbf{2}$ mS. Check: $2\sqrt{kI_D} = 2\sqrt{0.5 \times 2} = 2$ mS.

c. Unbypassed: $A_v = -\dfrac{(2)(4.7)}{1 + (2)(1)} = \mathbf{-3.13}$. Bypassed: $A_v = -(2)(4.7) = \mathbf{-9.4}$.

d. We need $R_D \parallel R_L = 5/2$ mS $= 2.5$ kΩ, so $R_L = \dfrac{1}{1/2.5 - 1/4.7} = \mathbf{5.34}$ kΩ.

6. Quick recall check

Question

1. $g_{m0} = 2I_{DSS}/|V_P|$ and $g_m = g_{m0}(1 - V_{GS_Q}/V_P)$. Only V_P in g_{m0} is taken as a magnitude.
2. $\sqrt{1/4} = 0.5$, so $g_m = 0.5\, g_{m0}$.
3. $Z_i = R_1 \parallel R_2$, $Z_o = R_D \parallel r_d$, $A_v = -g_m(R_D \parallel r_d)$.
4. $A_v = -g_m R_D/(1 + g_m R_S)$.
5. An n-channel D-MOSFET can be biased at $V_{GS_Q} > 0$, which makes $(1 - V_{GS_Q}/V_P) > 1$. A JFET's gate junction must not be forward biased, so $V_{GS} \le 0$.